



PRURIGO PIGMENTOSA

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Prurigo Pigmentosa (PP), also known as Nagashima Disease, is frequently referred to as the “keto rash” because there is a strong correlation between PP and the ketogenic diet. The ketogenic diet is a low-carbohydrate, high-saturated fat diet. PP is a rare inflammatory dermatosis characterized by pink itchy papules, macules, or vesicles, which evolves into brown spots in a net like pattern. PP predominately affects young females, likely due to higher prevalence of dieting behaviors as compared to other demographics. While people of all ethnicities can be affected, PP is most common among people of Asian descent.

The appearance of PP evolves over the course of the disease. The early stage of the rash typically appears with inflamed itchy papules or vesicles that eventually scale or crust. Over the course of weeks, these lesions spontaneously resolve and coalesce into a brown reticulated pattern. There is sparing of the mucous membranes, hair, and nails. Generally, PP lesions present in a symmetrical distribution with a strong predilection for the trunk and neck, but can also be found on the face, scalp, and in the suprapubic region. PP has many overlapping clinical features with other dermatosis and is often misdiagnosed as **urticaria, drug eruption, dermatitis herpetiformis, eczema, contact dermatitis, and confluent and reticulated papillomatosis**.

There is no definitive cause of PP, but various mechanical, hormonal, and metabolic triggers have been identified. These include ketosis, sweating, rubbing from clothing, chromium in acupuncture needles, **atopic dermatitis**, diabetes mellitus, anorexia nervosa, rapid weight loss, pregnancy, Sjogren’s Syndrome, and *Helicobacter pylori* (*H. pylori*) infection. Ketosis may be precipitated by a ketogenic diet, fasting, insulin-dependent diabetes mellitus, or bariatric surgery. When the rash is caused by the ketogenic diet, the timing of the onset of the rash is variable – six days to four months after initiation of the diet. In order to accurately diagnose PP, it is important to review underlying predisposing factors when taking the patient’s history.

The histopathologic features of PP depend on the timing of the biopsy. Early stages show superficial perivascular neutrophilic infiltrate, and it is postulated that ketone bodies induce this pattern of inflammation. Later stages are characterized by spongiosis and necrotic keratinocytes, followed by eosinophilic and lymphocytic infiltrate with melanophages in the papillary dermis

If a patient is found to have a rash that matches the pattern of PP and the rash is attributed to the ketogenic diet, patients should discontinue the diet. Resolution of ketosis is essential to remit PP, otherwise it may recur. First-line treatments include **tetracyclines**, including oral **minocycline** or **doxycycline**, which inhibit neutrophil chemotaxis and function and theoretically stop formation of the dermatosis. The use of **topical** and **systemic corticosteroids** are not effective at resolving PP.

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